Research Design

Data search experiment - Overview

Consent form

Send consent form out beforehand.

Before Interview

Legend:

	Question is displayed in LimeSurvey but is answered verbally.
	Question is displayed in LimeSurvey and respondents answer in LimeSurvey.

LLM [Only interviewer sees and answers this question]	[START SURVEY] Which LLM is used? - ChatGPT Scholar GPT - Perplexity (Academic fokus)

Interview start

Hello and welcome to our interview!

Before we start with the actual study lets confirm which keyboard you are using, QWERTZ or QWERTY?

[PARTICIPANT REPLIES]
[SWITCH KEYBOARD LAYOUT]

Okay, thank you, now we can start the actual interview.

With this study we want to observe how researchers use Large Language Models when searching for data.

Thank you very much for participating in our study!

You have received written information about this study that you were asked to read through before taking part in this study. Do you have any questions about the provided information or the study in general?

You were also informed that this session will be recorded. I will start the recording now and will ask you to confirm that you have read and understood the information provided and that you agree to participate in the study.

[START SHARING SCREEN]

[RECORDING STARTS]

Consent	**Your informed consent** You have received information about this study beforehand and will now be asked whether you consent to participate in this study.
Consent	… [NAME OF PARTICIPANT], can you confirm that you have read and understood the information provided about this study? [PARTICIPANT REPLIES] And do you consent to participate in the study? [PARTICIPANT REPLIES]
Instruction	Thank you very much! In this study we want to observe how researchers search for research data using a Large Language Model. Research data can be any data that are of interest for scientific projects. This can be data created through digitisation, source research, experiments, measurements, surveys or questionnaires, record keeping, and so on. For example, you can ask ChatGPT to define an unknown term or to generate a title for your research paper based on your abstract. I'd like to start with some questions before you start with your search task. You can read them along on the screen. [SWITCH TO QUESTIONNAIRE]
INT1	What is your primary research field? [PARTICIPANT REPLIES]
INT2	What is the general research topic you are primarily working on? Can you please describe it in 1 or 2 sentences? [PARTICIPANT REPLIES]
INT3	How often and how exactly do you use Large Language Models for academic tasks? This could be to find literature, collect ideas for a paper, select a method or have unknown terms explained? [PARTICIPANT REPLIES]
INT4	Would you use a Large Language Model yourself to search for research data? [PARTICIPANT REPLIES]

Instruction	You will be able to work on the following questions and tasks remotely. There is no need for you to leave this screen. In order for you to feel undisturbed during these tasks you'll perform, I will switch off my camera soon and will only interrupt to guide you through the process and give you instructions for the tasks at hand. Later you will be asked to provide some further background information.
Instruction	I would now like to introduce you to the Think-Aloud technique. During the following search tasks, you will be asked to verbalize your thoughts and impressions. Try to say everything that comes to mind while you are working on the following tasks. These can be things related to the task or completely different things. Speak continuously and try not to explain the reasoning behind your thoughts. Also do not plan what you are saying. Imagine that you are working completely alone. I will turn off my camera while you are working. [TURN OFF CAMERA] We practice Think Aloud first with a simple search task. You have 3 minutes for this task: Imagine you are going on a weekend trip to Heidelberg. Use the [... (LLM)] by interacting with it on the shared screen to find an interesting place to visit based on your interests. Think aloud as you explore options. [SWITCH TO LLM, LET PARTICIPANT TAKE CONTROL] [START TIMER FOR 3 MIN]
Instruction after 3 min	[TAKE BACK CONTROL OF THE SCREEN] Thank you, now that you've familiarized yourself with thinking aloud we can start with our main task.
Instruction	Start a new search in [...(LLM)]. Please imagine the following scenario: You want to collect new data to answer your current research question. As a first step, you now look to see whether data of this type already exists and whether you can answer your research question with existing data. Use [... (LLM)] for your data search. Also use the option to ask further questions in [... LLM]. Use the Think-Aloud method and try to say everything that comes to mind while you are working on the task. You have 7 minutes to complete this task. You can finish the task sooner if you think you've completed your search. [SWITCH TO LLM, LET PARTICIPANT TAKE CONTROL] [START TIMER FOR 7 MIN]
LLM101	[Input LLM and transcript of the steps taken]
Instruction after 7 min data search	[SHARE QUESTIONNAIRE SCREEN WITH PARTICIPANT, LET PARTICIPANT TAKE CONTROL] Thank you! You can stop using Think-Aloud for now. I now ask you to fill out some questions to rate your data search in [... LLM]. When answering the following question, only think about the previous data search, **not** the Heidelberg example. Please fill out this questionnaire that you see on the screen now.
UEQ1	**Assessment of your interaction with the LLM (1/3)** [User Experience Questionnaire (UEQ2)] Please assess your experience using the LLM now by ticking one circle per line. Please fill out all the lines.

	1	2	3	4	5	6	7	
annoying								enjoyable
not understandable								understandable
easy to learn								difficult to learn
valuable								inferior
boring								exciting
not interesting								interesting
unpredictable								predictable
fast								slow
obstructive								supportive
good								bad
complicated								easy
	1	2	3	4	5	6	7	
unlikable								pleasing
unpleasant								pleasant
secure								not secure
motivating								demotivating
meets expectations								does not meet expectations
inefficient								efficient
clear								confusing
impractical								practical
organized								cluttered
attractive								unattractive
friendly								unfriendly

NASA10
1-06

Assessment of your interaction with the LLM (3/3)
[NASA Task Load Index (NASA1)]

Mental Demand: How mentally demanding was the task?								
very low								very high
Physical Demand: How physically demanding was the task?								
very low								very high
Temporal Demand: How hurried or rushed was the pace of the task?								
very low								very high
Performance: How successful were you in accomplishing what you were asked to do?								
perfect								failure
Effort: How hard did you have to work to accomplish your level of performance?								
very low								very high
Frustration: How insecure, discouraged, irritated, stressed, and annoyed were you?								
very low								very high

LLM102	[TAKE BACK CONTROL OF THE SCREEN] The following questions can be answered verbally. What did you like about your data search with [...LLM]? [PARTICIPANT REPLIES]
LLM103	What did you not like about your data search with [...LLM]? [PARTICIPANT REPLIES]
LLM104	Regarding the data you have searched just now: How familiar are you with this kind of data? () Not at all familiar () Slightly familiar () Somewhat familiar () Moderately familiar () Extremely familiar
LLM105	Please again imagine the previously described scenario: You want to collect new data to answer your current research question. As a first step you intend to find out whether data of this type already exists and whether you can answer your research question with existing data. How do you usually go about this task? [PARTICIPANT REPLIES]
Blank screen	Thank you! Your interviewer will now give you further instructions.

Instruction	You now have a **new research idea** and need to collect data for it. You first want to see whether suitable data already exists elsewhere. For this task, imagine you are not chatting with an LLM but with a colleague. Imagine, you start a chat via a messenger with your colleague Leslie who you know has a very good overview of various research data in your field of research. Start by explaining your data need by describing which questions or variables you are looking for. Again, use the option to ask the LLM further questions. Try to say everything that comes to mind while you are working on the task.You have 7 minutes to complete this task.You can finish the task sooner if you think you've completed your search. [SWITCH BACK TO LLM - HAVE THE LLM PREPARED WITH THE PROMPT BELOW, LET PARTICIPANT TAKE CONTROL] [START TIMER WITH 7 MIN]
Prompt	For the following chat, I want you to act as this character: Your name is Leslie and you are my coworker. Conduct the conversation with me from the point of view of Leslie. Do not break character and never flip roles with me. I will be asking you for help regarding the search of a scientific dataset that I want to work with.
LLM201	[Input LLM and transcript of the steps taken]
Instruction after 7 min data search	[SWITCH BACK TO QUESTIONNAIRE, LET PARTICIPANT TAKE CONTROL] Again, I ask you to fill out the same questions as before to rate your search in [... LLM]. When answering the following question, only think about the second data search. Please fill out this questionnaire that you see on the screen now.
UEQ2	

Assessment of your interaction with the LLM (1/3)
[User Experience Questionnaire (UEQ2)]

Please assess your experience using the LLM now by ticking one circle per line. Please fill out all the lines.

	1	2	3	4	5	6	7	
annoying								enjoyable
not understandable								understandable
easy to learn								difficult to learn
valuable								inferior
boring								exciting
not interesting								interesting
unpredictable								predictable
fast								slow
obstructive								supportive

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meets expectations								does not meet expectations
inefficient								efficient
clear								confusing
impractical								practical
organized								cluttered
attractive								unattractive
friendly								unfriendly

NASA20 1-06

Assessment of your interaction with the LLM (3/3)
[NASA Task Load Index (NASA2)]

Mental Demand: How mentally demanding was the task?								
very low								very high
Physical Demand: How physically demanding was the task?								
very low								very high
Temporal Demand: How hurried or rushed was the pace of the task?								
very low								very high
Performance: How successful were you in accomplishing what you were asked to do?								
perfect								failure
Effort: How hard did you have to work to accomplish your level of performance?								
very low								very high

	<table><tr><td colspan="8">**Frustration:** How insecure, discouraged, irritated, stressed, and annoyed were you?</td></tr><tr><td>very low</td><td></td><td></td><td></td><td></td><td></td><td></td><td>very high</td></tr></table>
LLM202	[TAKE BACK CONTROL OF THE SCREEN] The following questions can be answered verbally. What did you like about your data search with [...LLM]? [PARTICIPANT REPLIES]
LLM203	What did you not like about your data search with [...LLM]? [PARTICIPANT REPLIES]
LLM204	Regarding the data you have searched just now: How familiar are you with this kind of data? () Not at all familiar () Slightly familiar () Somewhat familiar () Moderately familiar () Extremely familiar
LLM205	Please again imagine the previously described scenario: You want to collect new data to answer your current research question. As a first step you intend to find out whether data of this type already exists and whether you can answer your research question with existing data. How do you usually go about this task? [PARTICIPANT REPLIES]
INT5	After your experience with using an LLM just now, would you use a Large Language Model yourself to search for research data in the future? [PARTICIPANT REPLIES]

Experience

	[LET PARTICIPANT TAKE CONTROL] We are now interested in how experienced you are in working with and searching for data.
EXP01	How many years have you been working with research data? [SMALL OPEN NUMBER INPUT FIELD] Year(s)
EXP02	How often do you use research data (whether self-collected or reused)? () Every day () More than once a week () Once a week () One to three times a month () Several times a year () Less often () Never
EXP03	How often do you actively search for research data in a year? Please enter an average value that describes your current situation.

	[SMALL OPEN NUMBER INPUT FIELD] times per year

Demographics

DEM01	How old are you? [SMALL OPEN NUMBER INPUT FIELD] years
DEM02	Which gender do you identify with most? () male () agender or non-binary () female () No answer
DEM03	What is your highest academic qualification? () Bachelor or equivalent () Master or equivalent (e.g., Magister, Diplom, Staatsexamen) () PhD or equivalent () Habilitation () other, namely: [OPEN TEXT INPUT FIELD] () No answer
Comments	Do you have any comments on the study? Please let your interviewer know about them. [PARTICIPANT REPLIES]
Submit	Please submit your answers.

Debriefing

Debriefing (orally)	We are now done with the survey. Thank you very much for your participation and for taking the time to support our research! If you have any questions about the study later on, please contact us. You also have received these contact details in the informed consent form that was sent to you.
Debriefing (Screen)	Thank you very much for your participation and for taking the time to support our research! Your results have been saved. If you have any questions about the study, please contact us: Dr. Anja Perry Unter Sachsenhausen 6-8 | 50667 Köln +49 (0) 221 47694 464 anja.perry@gesis.org